Collectively, these applications underscore the transformation of drones from support tools to autonomous combat assets, central to next-generation warfare doctrines and multi-domain operations (MDOs).

5.5.2 Scientific and environmental applications

In the scientific and environmental domain, fourth-generation drones have become essential instruments for data-driven research, climate monitoring, and ecological management. Their AI-based sensing and edge computing capabilities allow for high-resolution data collection and real-time analysis in challenging environments.

- *Climate and atmospheric studies*: UAVs such as NASA's Global Hawk variant and India's CATS UAV collect atmospheric data for weather prediction, greenhouse gas measurement, and storm tracking, improving climate modeling accuracy.
- *Disaster monitoring and response*: Equipped with thermal imaging and LiDAR, drones support post-disaster mapping, flood risk assessment, and wildfire surveillance, offering immediate situational data for emergency management.
- *Environmental conservation*: UAVs are deployed for forest canopy analysis, anti-poaching operations, wildlife tracking, and deforestation mapping, enabling sustainable resource management.
- *Oceanographic and geological exploration*: Fourth-generation UAVs collaborate with autonomous underwater vehicles (AUVs) to perform coastal mapping, coral reef monitoring, and seismic data collection, demonstrating multi-domain research interoperability.

By enhancing precision and reducing human intervention, drones serve as vital instruments in promoting environmental stewardship and supporting